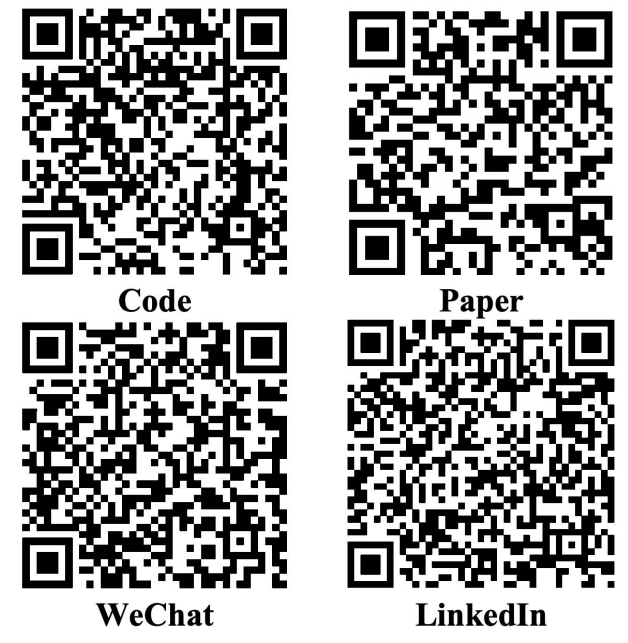


CaST-POI: Candidate-Conditioned Spatiotemporal Ranking for Next POI Recommendation

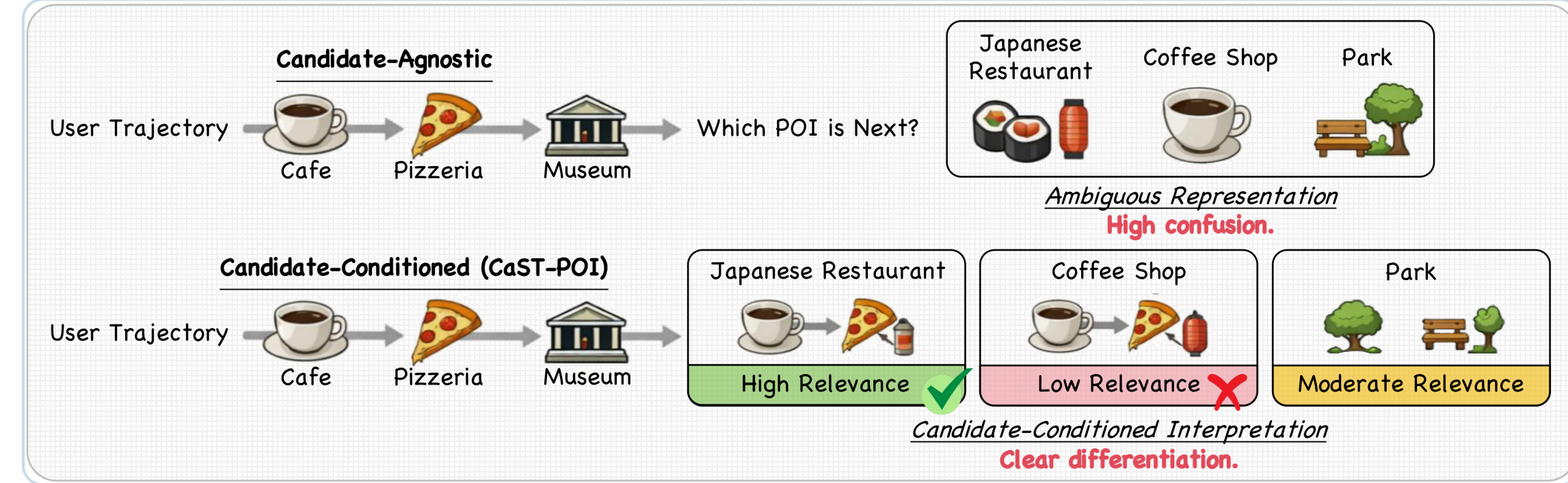
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1 Motivation

Most sequential rankers compress a trajectory into **one** user representation and score **every** candidate with it, ignoring where each candidate is.

Which past visits matter **depends on the candidate's location**: weekday-evening visits near the office are informative for a restaurant there, while a candidate near home makes a different subset of the same history relevant.

CTR-style **target attention** conditions the representation on the scored item, yet was built for catalogue items with **no canonical distance** to past behaviour.



Candidate-agnostic ranking yields one ambiguous representation; CaST-POI re-reads the same history per candidate.

Key idea: let every candidate **read the same history differently**, using **explicit geographic distance** to that candidate rather than relations inferred from item embeddings alone.

MRR gain over the strongest baseline on each dataset

+5.1%

NYC
not significant

+7.6%

TKY
p<0.001

+14.5%

CA
p<0.001

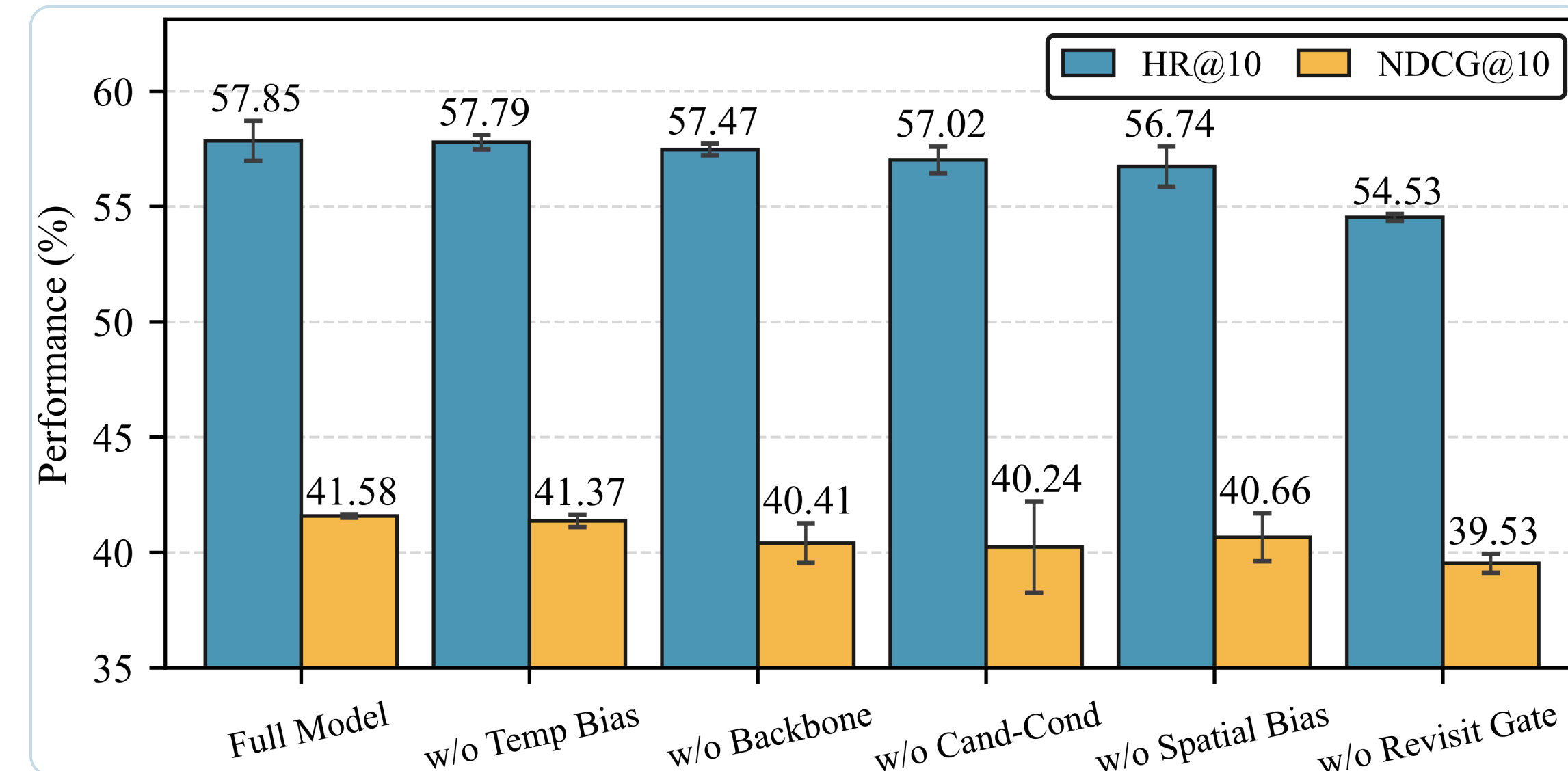
2 Contributions

A candidate-conditioned view of next-POI ranking. Ranking becomes a candidate-dependent representation problem, and missing geographic structure is the limitation of CTR-style target attention.

Spatiotemporally grounded target attention. Temporal and candidate-relative spatial biases enter the attention logits; one shared history encoding and few extra parameters keep **single-stage full-vocabulary** scoring without multistage retrieval or LLM reasoning.

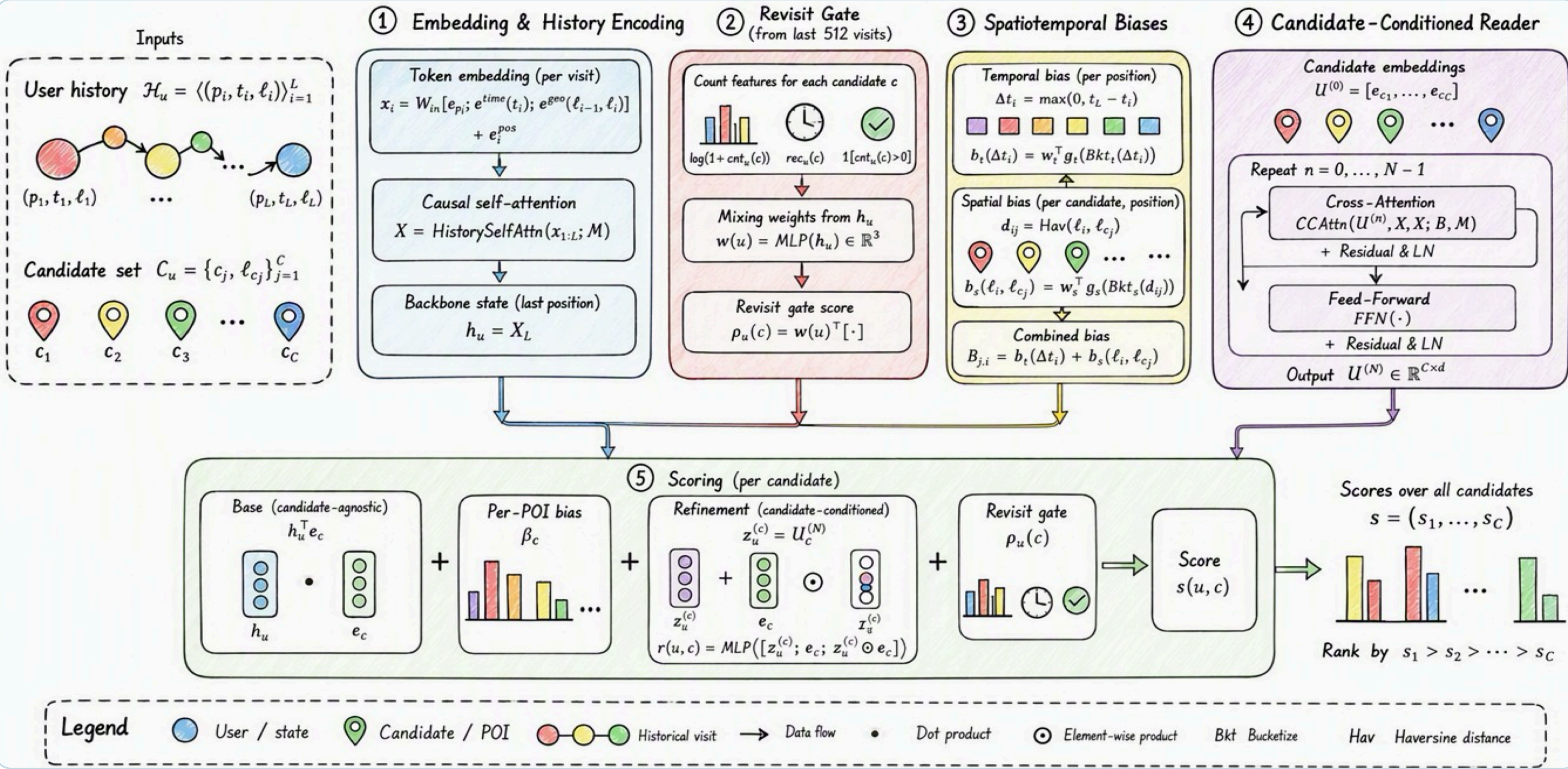
A controlled evaluation. Identical leave-one-out splits, full-vocabulary ranking, three seeds, paired tests with multiple-comparison correction, reported **null results**, and a parameter-free revisit heuristic.

8 Component Ablation (NYC)



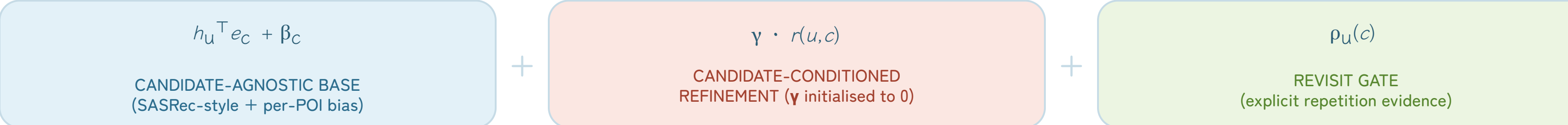
HR@10 drops on CA: spatial bias ~ 2.88 , candidate conditioning ~ 3.04 , revisit gate ~ 1.66 . On NYC / TKY the **revisit gate** dominates (~ 3.32 / ~ 3.07 , $p < 0.001$). The candidate-independent **temporal bias** shows **no measurable effect** in any of the 15 dataset-metric combinations.

3 The CaST-POI Framework



The trajectory is encoded **once** (1); candidate queries then read it through cross-attention (4), with logits shifted by the candidate-relative spatial and temporal biases (3). The four-term score adds the revisit gate (2), and all candidates are ranked in one stage (5).

4 Scoring Function & Components



Starting from a base scorer at least as expressive as SASRec isolates the **marginal** contribution of candidate conditioning; $\gamma = 0$ at initialisation removes that branch from the first optimisation step.

Encode the history once

Each visit concatenates POI, time (hour/day sinusoids + 4 slots) and **geo-displacement** features, projected to $d = 128$ and passed through causal self-attention. Keys/values are cached and reused by every candidate.

Spatial bias (where candidacy enters)

Haversine distance from each history location to the **candidate**, bucketed at $\{0.5, 2, 5, 20, 100\}$ km (walking $\rightarrow$ intercity), embedded and projected to a scalar added to the attention logits.

Cost. One history encoding plus chunked candidate scoring (1,024 at a time, scores identical to unchunked), $O(NCLd)$ per instance and **1.13-1.73M parameters**. Trained with full-vocabulary cross-entropy (label smoothing) + hard-negative margin and up-weighted *explore* instances on a single A100.

Candidate-conditioned reader

Candidates are queries over history keys/values: $\text{softmax}(\mathbf{QK}^T / \sqrt{d_h} + \mathbf{B} + \mathbf{M}) \mathbf{V}$, $N = 2$ blocks, 4 heads. The output feeds an MLP on $[z; e_c; z \odot e_c]$.

Temporal bias & revisit gate

Recency Δt bucketed at $\{1h, 6h, 24h, 7d, 30d\}$, shared across candidates. The gate mixes count, recency and visited-flag features over the last **512** visits with user-conditioned weights.

5 Datasets & Protocol

Dataset	#Users	#POIs	#Check-ins	Avg. Len.	#Test
NYC	1,047	4,974	101,753	97.2	1,044
TKY	2,280	7,832	403,146	176.8	2,280
CA	3,956	9,689	221,717	56.0	3,956

Per-user **leave-one-out** split over the released Foursquare (NYC, TKY) and Gowalla (CA) files; a content hash is verified whenever a model loads it, so every row below uses the identical split. Seven RecBole baselines are **retrained by us** with three seeds each, and every target is ranked against the **full vocabulary** (4,974 / 7,832 / 9,689 POIs). Significance: paired per-instance Wilcoxon signed-rank with Holm correction (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

6 Main Results

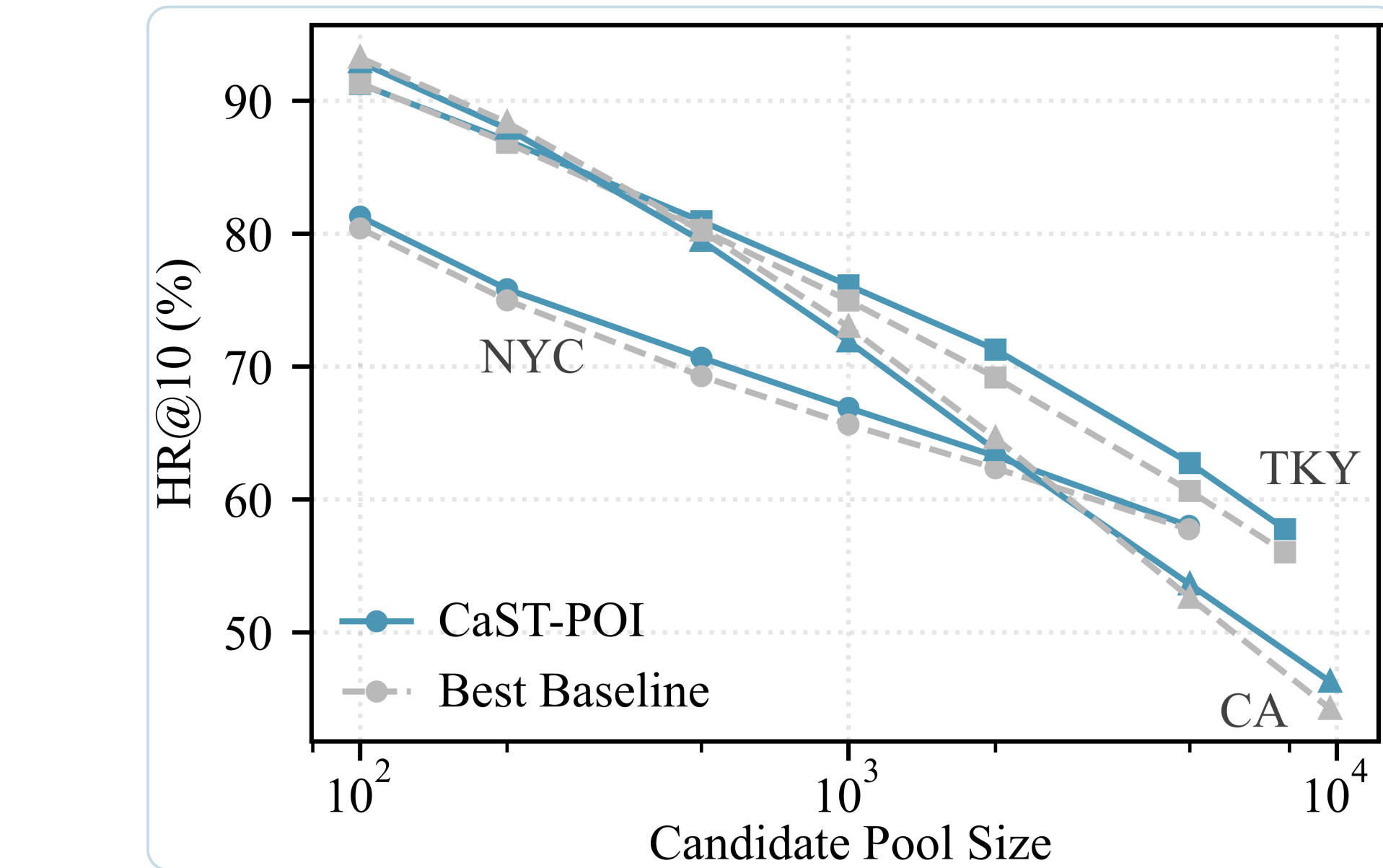
FULL-VOCABULARY RANKING (%)						
Method	NYC		TKY		CA	
	HR@10	MRR	HR@10	MRR	HR@10	MRR
SASRec	56.99 ± 10	35.35 ± 17	56.02 ± 41	32.12 ± 67	44.25 ± 23	23.69 ± 08
BERT4Rec	54.69 ± 25	33.78 ± 13	53.67 ± 98	31.96 ± 84	36.80 ± 48	21.37 ± 34
Caser	54.15 ± 06	33.12 ± 17	51.81 ± 46	31.69 ± 10	34.34 ± 48	19.88 ± 25
NARM	55.87 ± 22	34.55 ± 27	53.55 ± 08	<u>32.57± 34</u>	40.09 ± 48	23.11 ± 29
SR-GNN	52.81 ± 11	30.90 ± 27	53.49 ± 53	32.33 ± 41	38.29 ± 55	22.22 ± 25
STAMP	46.33 ± 21	30.51 ± 64	49.82 ± 29	31.40 ± 40	34.50 ± 11	20.80 ± 45
CORE	<u>57.76± 17</u>	<u>35.38± 16</u>	53.48 ± 89	27.58 ± 27	39.92 ± 25	22.57 ± 29
Revisit heuristic	54.60	26.94	41.54	18.97	35.52	20.05
CaST-POI	58.01 ± 15	37.20 ± 16	57.76 ± 75	35.05 ± 22	46.29 ± 29	27.12 ± 36
Improv.	+0.43%	+5.14%	+3.11%*	+7.61%***	+4.61%**	+14.48%***

Mean $\pm$ s.d. over three seeds; underline = strongest baseline per column, used for *Improv.* Over the full five-metric table CaST-POI is best in **14 of 15** dataset-metric combinations (only HR@5 on NYC is lower, by 0.29 points). Improvements are larger for position-sensitive metrics, so candidate conditioning mainly **reorders** what a strong ranker already retrieves. NYC is reported as a **null result**: no NYC improvement survives Holm correction under the Wilcoxon test, although a paired bootstrap does reject for NDCG@10 and MRR.

7 Where CaST-POI Sits

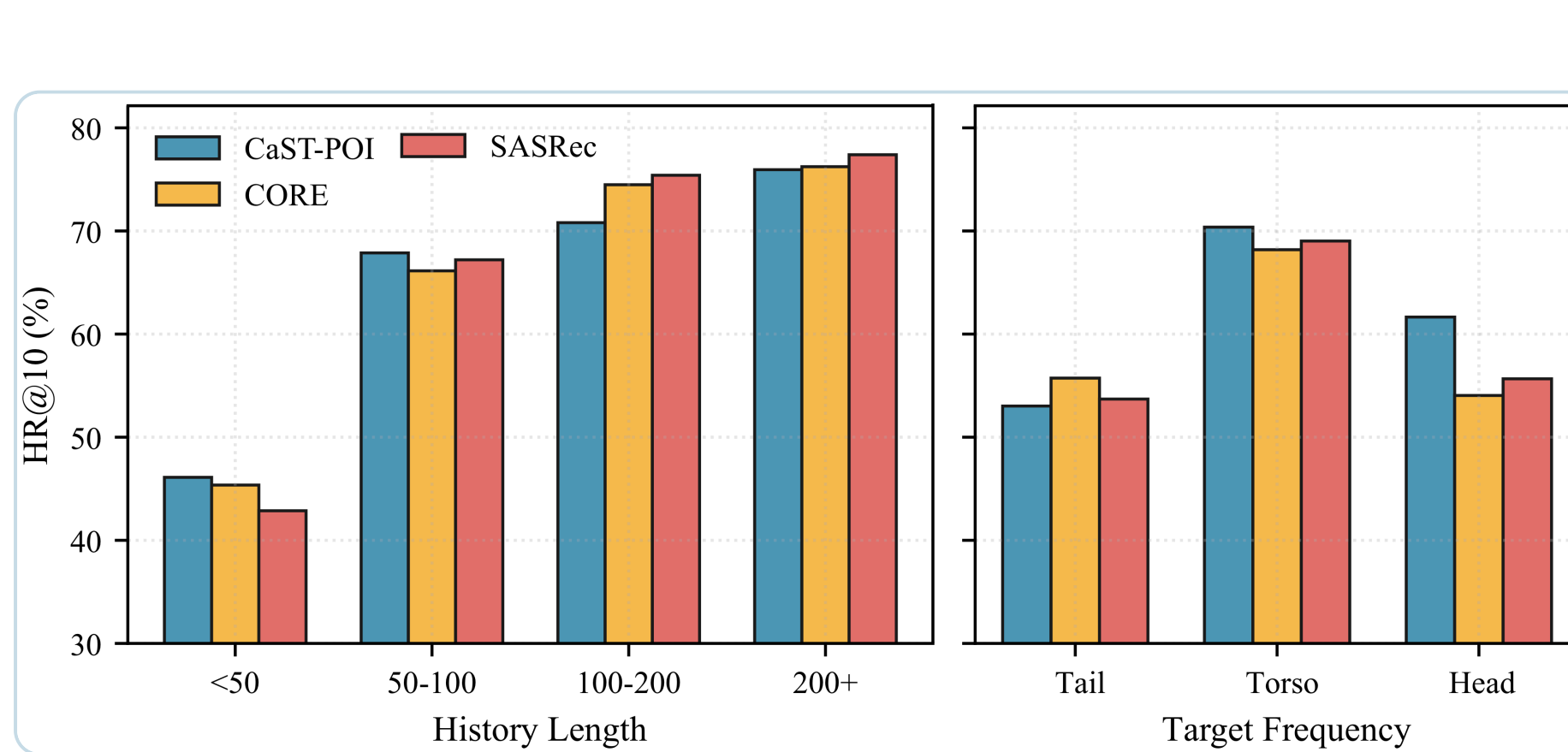
Family	Candidate-conditioned	Explicit geometry
Sequential / POI rankers SASRec, CORE, GETNext, STAN	✗	trajectory-level
CTR target attention DIN, DIEN, BST, SIM	✓	✗
LLM / agent POI pipelines LLM4POI, CoMaPOI, ROTAN	pipeline level	✗
CaST-POI (ours)	✓	candidate-relative

9 Candidate Pool Size



Sampled-pool scores are computed in closed form from full-vocabulary ranks. At $N = 100$ the ordering **reverses** on CA (SASRec 93.24 vs. 92.91); full ranking gives 44.25 vs. **46.29** ($p < 0.01$). TKY behaves likewise.

10 Robustness (NYC)



CaST-POI leads in the short-history groups on NYC and in **every** group on TKY and CA: for users with at least 200 visits it reaches 70.75 against SASRec's 69.03 on TKY, and 57.04 against 56.19 on CA. Gains concentrate on **head and torso** POIs (+5.99 points on the NYC head group), so the method is **not** a cold-start solution. On the NYC head group the parameter-free revisit heuristic still wins with 64.08, a reminder of how repetitive these benchmarks are.

11 Takeaways

- Geometry, not more sequence modelling.** Candidate-relative distance inside the attention logits provides the gain: largest on state-scale CA (MRR +14.5%), smallest in dense NYC.
- Explicit revisit evidence matters** where repetition is high: 64.1% (NYC) and 68.5% (TKY) of targets are revisits, against 44.7% on CA.
- A reported null result.** The candidate-independent temporal bias yields no measurable improvement; recency is already available to the causal backbone.
- Evaluate on the full vocabulary.** 100-candidate sampling inverts the method ordering on two of the three datasets.
- Cheap and single-stage.** 1.1-1.7M parameters, one history encoding, no LLM and no multistage retrieval.
- Reproducible by construction.** Hash-verified splits shared by every method, baselines retrained here, three seeds, and code plus split scripts released.